IISER Mohali

[January 2026 Session]

PHY306 (Advanced Quantum Mechanics)

Midsem-I

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Max. Marks : 20

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1. Compute the first-order corrections to the wavefunctions and the energies of the bound states of an infinitely deep square well between $x = 0$ and $x = a$, in presence of the perturbation
$$\delta H = \kappa\, \delta\Big(x - \frac{a}{2}\Big),\ \ \kappa = \frac{\pi^2\hbar^2}{4ma}.$$
Which states are unaffected by the perturbation, and why? Write the ground-state wavefunction including the first-order correction in its most simplified form. [**5**]

2. Using the fine structure correction result for hydrogen atom,
$$E_{fs} = E_n^0\left[1 + \frac{\alpha^2}{n^2}\left(\frac{n}{j+\frac{1}{2}} - \frac{3}{4}\right)\right],$$
draw the splitting of $n = 3$ energy levels and order them in energy. Label them with appropriate spectroscopic notation and state the degeneracy of energy levels before and after the splitting. [**5**]

3. State True/False with justification for the following statements. [**10**]
   (a) Fine structure splitting in deuterium is slightly larger than in hydrogen.
   (b) Due to hyperfine interaction, the degeneracy of $1s_{1/2}$ fine-structure level, including nuclear spin, is completely lifted.
   (c) If the degenerate subspace exhausts the Hilbert space, then the first order correction $|n^{(1)}\rangle = 0$.
   (d) The harmonic oscillator potential can be treated as perturbation to the free particle Hamiltonian in the sense required by perturbation theory.
   (e) The spin-orbit interaction breaks the rotational symmetry in the hydrogen atom.

Inputs

1. Infinite square well between $x = 0$ to $x = a$

$$\psi_n^0(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \qquad E_n^0 = \frac{n^2\pi^2\hbar^2}{2ma^2}, \quad n = 1, 2, 3, \ldots$$

2. Time-independent perturbation theory

$$H = H^0 + \delta H$$

$$|n^{(1)}\rangle = -\sum_{m \neq n} \frac{\langle m|\delta H|n\rangle}{E_m^0 - E_n^0}\ |m\rangle$$

3. Hydrogen atom

$$E_n^0 = -\mu c^2 \frac{\alpha^2}{2n^2} = -\frac{13.6}{n^2}\ \text{eV}$$